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Cyse 300

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The importance of Mitigating Cybersecurity Breach

BLUF: In 2019, Capital One experienced a data breach caused by Paige Thompson, a former Amazon employee who gained unauthorized access to Capital One software by exploiting existing vulnerabilities. As a result, the cybercriminal was able to steal sensitive information stored within the business software, placing multiple employees and clients at risk.

Having mitigation procedures is critical when running a business because, if a cybersecurity breach occurs, both employees and business clients are at risk of having their information leaked or facing other dangerous attacks. Because of this, businesses that have assets such as money, digital information, and employee access control should implement proper security practices, especially in 2026. With the growth of cybersecurity as a discipline, cybercriminals and new attack methodologies may also be on the rise. Throughout this research, I will examine the security breach involving Capital One software that occurred in 2019, which led to the theft of information from multiple individuals. Furthermore, this paper will explain the vulnerabilities involved, how the criminal exploited the vulnerability, the aftermath of the security breach, and what security measures should have been implemented to mitigate the cyberattack.

A vulnerability that was present was Paige Thompson herself, as she acted as an insider threat due to her prior employment with Amazon. In addition, she was able to use Amazon Web Services (AWS) to identify improperly configured firewalls, which is how she discovered Capital One's firewall vulnerability. Another critical vulnerability was the improper installation of Capital One's firewall. As stated in the article *Be an Expert: A Critical Thinking Approach to Responding to High-Profile Cybersecurity Breaches*, "Thompson created scanning software that identified AWS customers with improperly configured firewalls and was able to access the systems of more than 30 companies with improperly configured firewalls, including Capital One" (Boss 2024). This statement explains how Paige Thompson was able to bypass Capital One's firewall, leading to unauthorized access to sensitive information and the installation of software used to mine cryptocurrency. This vulnerability could have been mitigated if Capital One had established proper procedures to securely install firewalls and regularly patch their software to prevent unauthorized access.

Although Paige Thompson was arrested before she could sell the stolen information, the Capital One breach still had serious repercussions. The incident exposed sensitive customer data, damaged public trust, and highlighted weaknesses in Capital One's cloud security practices. While the stolen data was not sold, the loss of confidentiality placed customers at long-term risk of identity theft and financial fraud. While some cyber incidents do not involve direct system intrusion, such as the Advocate Health case in which physical devices containing patient information were stolen, the outcome is similar, illustrating that once sensitive information is exposed, security can no longer be guaranteed. According to the article *Trends and Preventive Strategies for Mitigating Cybersecurity Breaches in Organizations*, stolen data often includes personal identifiers such as names, addresses, and Social Security numbers (Astani 2016). The

Capital One breach demonstrates how cybersecurity incidents can result in lasting reputational damage and emphasizes why cybersecurity should be a priority and incorporated into organizational training.

In conclusion, the Capital One security breach conveys the importance of implementing strong cybersecurity mitigation practices. This incident demonstrates the need to ensure that proper procedures and software are correctly installed within business systems, as improper cybersecurity practices can be exploited by insider threats, resulting in unauthorized data leaks that endanger both employees and clients. Bringing awareness to cybercrime is especially important in employee training, as increased knowledge improves an organization's ability to mitigate cyber threats and protect financial institutions. If these actions are taken, businesses will be better equipped to safeguard their assets and clients.

Works Cited

Boss, S. R., Gray, J., & Janvrin, D. J. (2024). Be an Expert: A Critical Thinking Approach to Responding to High-Profile Cybersecurity Breaches. *Issues in Accounting Education*, 39(1), 93–121. <https://doi.org/10.2308/ISSUES-2021-094>

“TRENDS AND PREVENTIVE STRATEGIES FOR MITIGATING CYBERSECURITY BREACHES IN ORGANIZATIONS.” *Issues in Information Systems*, 2016, https://doi.org/10.48009/2_iis_2016_208-214.